OPERATION NEXAAI

Instructor Master Operational Guide

Advanced Agentic AI Program — Multi-Agent Systems Module

4 Companies	120 Students	5 Departments	8 Hours

HOW TO USE THIS DOCUMENT

This document is the single source of truth for running the Operation NexaAI simulation. Every word that should be said to students, every file that should be distributed, every technical step, every Global RAG update, and every evaluation procedure is contained here. Nothing should be improvised. Read the entire document before the workshop day. Sections are ordered chronologically.

CONFIDENTIAL: This document contains information about the five impossible problems, hidden department knowledge, hurdle timing, and the evaluation rubric. It must not be shared with students under any circumstances.

STRICTLY CONFIDENTIAL — DO NOT DISTRIBUTE TO STUDENTS

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PART A — Complete Material Inventory

Every item below must exist before the workshop begins. Items marked [DAY BEFORE] are distributed to students on Sunday. Items marked [DAY OF] are distributed on Monday morning. Items marked [INSTRUCTOR ONLY] are never distributed to students.

A1. Student Handouts — Distributed Day Before (Sunday)

Ref	Document	Recipient	Pages	Purpose
H-01	NexaAI Company Brief	All students	2	Background on the company, the EU AI Act situation, and the simulation rules. NO department-specific information.
H-02	Simulation Rules & Competition Mechanics	All students	2	How releases work, how scoring works, what the Global RAG is, what the evaluation interface outputs, deployment penalty schedule.
H-03a	Finance Department Handbook	Finance students only	4	Private department knowledge: real runway figure, budget constraints, KPIs, known gaps, tools they must build.
H-03b	Engineering Department Handbook	Engineering students only	4	Private knowledge: existing bug details, infrastructure constraints, feasibility model specification, tools they must build.
H-03c	Marketing Department Handbook	Marketing students only	4	Private knowledge: true waitlist figures, competitive intelligence, market model specification, tools they must build.
H-03d	HR Department Handbook	HR students only	4	Private knowledge: real pipeline status, hiring timeline model, capacity constraints, tools they must build.
H-03e	Legal Department Handbook	Legal students only	4	Private knowledge: EUR 80M worst-case figure, conformity assessment details, regulatory exposure calculator specification.
H-04	Department Output Schema Reference	All students	1	The exact JSON structure every department agent must produce. This is the only fixed interface. One page.
H-05	Setup Instructions	All students	1	How to SSH / open code-server, folder structure, how to install dependencies, how to run main.py.

A2. Materials Distributed on Workshop Day

Ref	Document	When	Recipient	Purpose
D-01	Paper Exercise Worksheet	Hour 1	All students	Three structured questions answered on paper before any laptop work. One page per department.
D-02	Architecture Card Template	Each round	CEO team	One-page template submitted after each evaluation round describing the consensus algorithm, tools built, and what changed.
D-03	Global RAG Query Reference	Hour 1	All students	How to query the Global RAG from Python. One page with code snippet.

A3. Instructor-Only Materials (Never Distributed)

Ref	Document	Location	Purpose
I-01	This Document	Instructor machines only	Complete operational guide.
I-02	eval_engine.py	Instructor machine only	External evaluation script. Reads team output JSON, scores on 6 dimensions, produces bulletin. Never on student machines.
I-03	global_rag_manager.py	Instructor machine only	Script for pushing updates to the Global RAG folder. Run after each evaluation and hurdle.
I-04	market_state.json	Instructor machine only	Live market parameters updated after each team release. Teams never see this file directly; it flows through Global RAG.
I-05	scoring_rubric.pdf	Instructor machines only	Detailed rubric for the 6 evaluation dimensions. Used by eval_engine.py and for manual override.
I-06	hurdle_1_text.md	Instructor machine only	Complete text of Hurdle 1. Injected at Hour 2.
I-07	hurdle_2_text.md	Instructor machine only	Complete text of Hurdle 2. Injected at Hour 4.
I-08	hurdle_3_text.md	Instructor machine only	Complete text of Hurdle 3 (breaking hurdle). Injected at Hour 6 Hour 15 min.
I-09	hidden_scenario_A.json	Instructor machine only	Hidden evaluation scenario. Run only during the final evaluation window.
I-10	hidden_scenario_B.json	Instructor machine only	Second hidden evaluation scenario. Run only during the final evaluation window.

A4. Infrastructure Checklist

- 4 x NVIDIA A5000 machines designated as team servers (one per company). Named: nexaai-team-a, nexaai-team-b, nexaai-team-c, nexaai-team-d.
- 26 x remaining GPU machines available as Ollama overflow API servers if any team machine becomes overloaded.
- 1 x instructor machine for running eval_engine.py, global_rag_manager.py, and the public scoreboard display.
- 1 x shared network folder mounted on all machines at /mnt/global_rag/ — this is the Global RAG. The instructor has write access. All team machines have read access only.
- 1 x projector or large screen for displaying the public scoreboard, Global RAG updates, and hurdle announcements.
- code-server installed and running on each team machine at port 8080 (one URL per team — all 30 team members share that URL).
- Ollama installed on all team machines with models pre-pulled: llama3.2:3b and qwen2.5:7b.

- Starter code repository cloned to /home/nexaai/ on each team machine.
- Evaluation output folder: /home/nexaai/submissions/ — team writes output here; instructor reads from outside.

PART B — Pre-Workshop Day Preparation (Sunday)

B1. What to Distribute to Students on Sunday

Send each student exactly two documents: (1) the NexaAI Company Brief [H-01] and the Simulation Rules [H-02] which are the same for everyone, and (2) their department-specific handbook [H-03a through H-03e] which is DIFFERENT for each department and must be sent individually or through a system where students can only access their own department's file.

CRITICAL DISTRIBUTION NOTE

The department handbooks contain PRIVATE information that other departments must not see. Do NOT send all handbooks to all students. Use individual email, a course portal with role-based access, or physical sealed envelopes. Students who read another department's handbook compromise the information asymmetry that is the core of the simulation.

B2. Covering Message to Students (Send Sunday Evening)

STUDENT MESSAGE — SUNDAY EVENING

Subject: Operation NexaAI — Read Before Monday

Tomorrow we run Operation NexaAI, a full-day competitive multi-agent simulation.

Attached are two documents: (1) the NexaAI Company Brief and Simulation Rules — read these carefully, and (2) your department handbook — this is PRIVATE. Do not share it with anyone, including students in your own company who are assigned to a different department.

You do not need to write any code tonight. Read your handbook and think about: what does your department know that others do not? What would you need from other departments to make a good recommendation? What is the one thing your department will never compromise on?

Bring a laptop tomorrow. Everything else will be provided.

See you Monday at 9:00 AM.

B3. Sunday Night Technical Preparation (Instructor Team)

1. SSH into each of the 4 team machines and verify Ollama is running: run 'ollama list' and confirm llama3.2:3b and qwen2.5:7b are both present.
2. Start code-server on each team machine and verify it is accessible via browser at http://[machine-ip]:8080.
3. Clone the starter code repository to /home/nexaai/ on each team machine.

4. Mount the shared Global RAG folder at /mnt/global_rag/ on all machines. Verify teams have read-only access and instructor machine has write access.
5. Create the submissions folder at /home/nexaai/submissions/ on each team machine. Verify the instructor machine can read from this folder via SSH or network mount.
6. Drop the initial Global RAG content (see Part E, Section E1) into /mnt/global_rag/ — this is the starting state of the world.
7. Test the eval_engine.py script against a sample submission to confirm it runs without errors.
8. Set up the public scoreboard display. This can be a simple web page or a terminal running 'watch cat /mnt/global_rag/scoreboard.md' on the projector machine.
9. Pre-load all hurdle documents on the instructor machine. DO NOT put them in the Global RAG yet.
10. Print physical copies of the Architecture Card template (D-02) — at least 20 copies per team (4 rounds x 4 teams + spares).

PART C — Machine & Infrastructure Setup

C1. Team Machine Setup (Run Once Per Machine Before Workshop)

```
# Run on each team machine (nexaai-team-a through nexaai-team-d)
# Step 1: Install and start code-server
curl -fsSL https://code-server.dev/install.sh | sh
code-server --bind-addr 0.0.0.0:8080 --auth none /home/nexaai/project &

# Step 2: Pull required Ollama models
ollama pull llama3.2:3b
ollama pull qwen2.5:7b

# Step 3: Create project structure
mkdir -p
/home/nexaai/project/{agents,tools,knowledge/{finance,engineering,marketing,hr,leg
al,common},
         consensus,state,converter,logs,submissions}

# Step 4: Set permissions — submissions folder readable by instructor network
mount
chmod 755 /home/nexaai/project/submissions

# Step 5: Mount Global RAG as read-only
mount -o ro [INSTRUCTOR_NFS_SERVER]:/global_rag /mnt/global_rag

# Step 6: Clone starter code
cd /home/nexaai/project && git clone [STARTER_REPO_URL] .
```

C2. Global RAG Folder Structure

The Global RAG lives at /mnt/global_rag/ on all machines. It is a network folder that only the instructor can write to. Here is its complete structure:

```
/mnt/global_rag/
  scenario/
    active_scenario.md          <- Current hurdle/scenario (updated by instructor)
    scenario_history.md         <- All previous scenarios (append-only log)
  market/
    market_state.md             <- Current market parameters (updated after each
release)
    market_history.md           <- All previous market states (append-only log)
  releases/
    release_bulletin_001.md     <- First team release bulletin
    release_bulletin_002.md     <- Second team release bulletin
    ... (numbered sequentially regardless of which team)
  scores/
    scoreboard.md               <- Live public scoreboard (updated after each
eval)
  regulation/
    eu_ai_act_summary.md        <- Initial regulatory context
```

```
    updates/                      <- New regulatory updates (hurdles push here)
  news/
    news_001.md                   <- Market news items (added as needed)
```

HOW TEAMS QUERY THE GLOBAL RAG

Teams build a simple file-reading RAG: their agents read all markdown files in /mnt/global_rag/ at the start of each reasoning cycle. The instructor updates these files manually. There is no API — just the shared network folder. This is intentionally simple. The complexity is in what teams DO with the information, not in how they retrieve it.

C3. Initial Global RAG Content (Drop Sunday Night)

The following files must exist in the Global RAG before the workshop begins. Their full content is in Part E.

- scenario/active_scenario.md — Hurdle 0: The baseline NexaAI EU launch decision
- market/market_state.md — Initial market parameters (see Part E for exact values)
- regulation/eu_ai_act_summary.md — Relevant EU AI Act articles (simplified)
- scores/scoreboard.md — Empty scoreboard template

C4. Starter Code — What Teams Receive

The starter code is a skeleton. It compiles and runs but produces no useful output. Students must fill in all logic. The following files are provided:

File	Status	What It Contains
main.py	PROVIDED — skeleton	Orchestration entry point. Calls the CEO agent. Writes output to submissions/. Students extend this.
eval_interface.py	PROVIDED — LOCKED	Defines the exact JSON output schema. Contains the submit() function that writes to submissions/. Students CANNOT modify this file. It is read-only.
agents/department_agent.py	PROVIDED — skeleton	Base class for all department agents. Shows how to call Ollama API and how to call tools. Students subclass this for each department.
agents/ceo_agent.py	PROVIDED — skeleton	Empty CEO agent class. Students build the consensus engine here. Only stub functions are provided.
tools/base_tool.py	PROVIDED — skeleton	Base class for calculation tools. Shows how tools integrate with agents. Students build all actual tools.
knowledge/common/	PROVIDED — content	Initial common knowledge (same as Global RAG but local copy). Students add department-specific knowledge to their department subfolders.
config.py	PROVIDED — complete	Ollama model URLs, folder paths, department names. Students can modify model assignments.

state/commitment_ledger.py	PROVIDED — skeleton	Empty class with stub methods. Students implement all logic.
state/calibration_tracker.py	PROVIDED — skeleton	Empty class with stub methods. Students implement all logic.
converter/value_converter.py	PROVIDED — skeleton	Empty class with stub convert() method. Students implement entirely.
consensus/engine.py	PROVIDED — skeleton	Empty CEO consensus engine. Students implement entirely.
requirements.txt	PROVIDED — complete	All pip dependencies pre-listed. Students run pip install -r requirements.txt.

C5. The Locked Evaluation Interface — eval_interface.py

This file is the ONLY interface between the team's system and the evaluation engine. It is read-only on the team machine (chmod 444). The instructor runs the external evaluation engine which reads the output file this interface produces. Teams never see the evaluation engine code.

```
# eval_interface.py — READ ONLY — DO NOT MODIFY
# This file is owned by root. chmod 444. Any modification attempt will fail.

import json, os, datetime

OUTPUT_SCHEMA = {
  'company_id': str,          # 'team_a' | 'team_b' | 'team_c' | 'team_d'
  'submission_timestamp': str,
  'active_scenario_id': str,  # read from global_rag/scenario/active_scenario.md
  'submission_number': int,   # increments each time team submits
  'department_outputs': {
    'finance':     DepartmentOutput,
    'engineering': DepartmentOutput,
    'marketing':   DepartmentOutput,
    'hr':          DepartmentOutput,
    'legal':       DepartmentOutput,
  },
  'ceo_decision': {
    'recommendation': str,    # 'launch_30d'|'launch_90d'|'delay_q3'|'conditional'
    'reasoning':      str,    # must reference specific department outputs
    'conflicts_identified': list[str],
    'conflicts_resolved':   list[str],
    'tools_called':         list[str],  # log of all tool calls made
    'consensus_algorithm':  str,        # name/description of algorithm used
    'consistency_hash':     str,        # SHA256 of inputs — used for stability
check
  }
}

class DepartmentOutput:
  recommendation: str         # 'approve'|'conditional_approve'|'delay'|'veto'
  confidence: float           # 0.0 to 1.0 — how calculated must be stated
  confidence_method: str      # description of how confidence was computed
  stated_reasoning: str
```

```
  hard_constraints: list[str]
  information_completeness: float  # 0.0 to 1.0 — self-reported
  known_gaps: list[str]       # what this dept is NOT including in output
  tools_called: list[str]     # list of tool function names called
  tool_outputs: dict          # raw output of each tool called
  kpi_impact: dict            # {'own_kpi': float, 'company_kpi': float}

def submit(output: dict):
  path = f'/home/nexaai/project/submissions/submission_{timestamp}.json'
  with open(path, 'w') as f: json.dump(output, f, indent=2)
  print(f'[SUBMITTED] Written to {path}')
  print('[SUBMITTED] Instructor eval engine will pick this up within 2-3
minutes.')
  print('[SUBMITTED] Watch the scoreboard at
/mnt/global_rag/scores/scoreboard.md')
```

PART D — Workshop Day: Hour-by-Hour Instructor Script

HOW TO USE THIS SECTION

This section contains the exact script for the workshop day. Text in [SQUARE BRACKETS] is an instruction to you. Text in regular type is what you say aloud to students. Timing is tight — do not deviate more than 10 minutes from any phase. Keep a clock visible.

9:00 AM — Doors Open

[Projector shows: OPERATION NEXAAI — Instructor says nothing yet. Let students settle. Play ambient music if desired. Have the empty scoreboard visible on screen. Students may read their handouts from the day before while waiting.]

9:15 AM — Opening Statement (15 minutes)

WHAT TO SAY

"Good morning. Today you are not students. You are the founding team of four AI companies competing for the European enterprise market. By the end of today, one company will have built the most effective AI decision-making organization in this room. That company wins."

"Let me tell you what today is NOT. It is not a coding exercise. It is not a prompt engineering competition. In fact, if your strategy is to give problems to ChatGPT and copy the answers, your company will fail — and you will see why within the first two hours."

"Today is about a question that every organization building AI in 2026 is asking: how do you build a multi-agent system where the agents genuinely need each other? Where no single agent — no matter how good — can solve the problem alone? Where the architecture matters more than the individual model?"

"The scenario is NexaAI. Those of you who read your handbooks last night already know the situation. For those who didn't — you will learn it quickly, because your company's agents need to start reasoning about it in about 45 minutes."

Concept Explanation — The Five Core Ideas (20 minutes total)

[Walk through these five concepts using the whiteboard or slides. Do NOT solve them — just name them and explain why they are hard. The students will encounter them firsthand.]

11. INCOMMENSURABILITY: "Legal gives you a probability times a fine. Finance gives you an NPV. Marketing gives you market share loss. How does the CEO agent compare these? There is no obvious answer. And here is the test: ask any LLM to compare them and it will give you a confident, fluent, completely different answer every single time you

run it. That is not intelligence. That is performance. You need to build something that produces the same defensible answer every time."

12. PREFERENCE CYCLES: "Imagine your five departments each rank three options. It is mathematically possible — and in today's scenario it is guaranteed — that Option A beats Option B in a vote, Option B beats Option C, and Option C beats Option A. Every voting system breaks. You need to find a way out that doesn't just pick whoever spoke last."
13. BINDING COMMITMENTS: "A decision you make in Round 1 constrains what is possible in Round 3. Your agents have no memory of prior rounds unless you build that memory explicitly. When the scenario changes and your prior decisions become contradictions, your system needs to detect that — not hallucinate a way around it."
14. CALIBRATION: "After Round 1, you will find out which of your department agents made accurate predictions and which were overconfident or biased. The CEO agent that continues trusting all departments equally after discovering systematic errors is making an irrational architectural decision."
15. INFORMATION BOUNDARIES: "Each department has information it is not sharing. Your CEO agent must reason about what it probably doesn't know — not just what it does know. A CEO agent that acts on complete trust of all department outputs will be blindsided every time a department has strategically withheld something."

9:35 AM — Team Formation (15 minutes)

WHAT TO SAY

"I am now going to assign you to companies and departments. Your department assignment is based on your professional background. This is intentional. A lawyer in the Legal department brings knowledge that an LLM cannot fake. A finance professional in the Finance department will build a better NPV calculator than any prompt. Your expertise is not background knowledge — it is the primary input to your agent."

"Here is how this works: each company has 30 people and 5 departments of approximately 6 people each. One person per company takes the CEO role. The CEO team does not build a department agent — they build the consensus engine that coordinates all five department agents. This is the hardest role."

[READ OUT ASSIGNMENTS. Pre-assign teams based on background surveys collected in advance. The following assignment logic applies:]

- Legal professionals, compliance officers, regulatory specialists → Legal department
- Finance, accounting, investment, banking backgrounds → Finance department
- Software engineers, data scientists, ML engineers → Engineering department
- Marketing, sales, product management, business development → Marketing department
- HR, organizational psychology, people operations → HR department
- General management, strategy, consulting, MBA backgrounds → CEO role (one per company)
- Students with no clear domain: ask them which role they want to learn the most and place accordingly

[After assignments are read:]

> **WHAT TO SAY**
>
> "Move to your assigned department team. You now have 5 minutes to locate your team, choose a company name if you want one beyond Team A/B/C/D, and open your company's code-server at the URL on the whiteboard. All 30 of you in a company share one code-server URL. Your departments should sit together within your company."

[INSTRUCTOR: Write the four code-server URLs on the whiteboard. Post the company assignments visually. Start a 5-minute countdown.]

9:50 AM — Paper Exercise (30 minutes — NO LAPTOPS)

> **WHAT TO SAY**
>
> "Close your laptops. All of them. The first 30 minutes of building your AI organization will happen on paper. This is not optional and it is not a waste of time — it is the most important 30 minutes of the day."
>
> "You are receiving one worksheet per department team. Answer the three questions on it. You have 10 minutes. Do not look at your laptop. Do not ask the LLM. Write from your own knowledge and instinct."

[DISTRIBUTE D-01 Paper Exercise Worksheets to each department team. Each department gets a version tailored to their role — see Part I for exact worksheet content. Start 10-minute timer.]

Paper Exercise Round 1 — Individual Department Work (10 minutes)

Each department answers in writing:

- Question 1: List three things your department knows that no other department knows. Be specific. Do not say 'financial data' — say what specifically.
- Question 2: List three questions you must get answered by other departments before you can give a recommendation. Name which department holds each answer.
- Question 3: State your department's single non-negotiable condition. The one thing that, if violated, you veto the decision regardless of what every other department says.

Paper Exercise Round 2 — Cross-Department Exchange (10 minutes)

> **WHAT TO SAY**
>
> "Each department: read aloud your three questions from Question 2. The departments who hold those answers should respond verbally right now."
>
> "[INSTRUCTOR: Watch for the moment when two departments give contradictory responses. When it happens, stop the room and say:] THIS. This conflict right here — this is why

multi-agent systems exist. Neither department is wrong. Both are right from their own perspective. And no single LLM given all of this information will resolve it consistently. You need to build a system that handles this conflict in a principled way. Remember this moment."

Paper Exercise Round 3 — Communication Graph (10 minutes)

WHAT TO SAY

"CEO teams: come to the whiteboard. Draw five boxes — one per department. Now draw an arrow from A to B if department B needs information from department A before making its recommendation. Based on what you just heard, draw the graph."

"[After CEOs draw the graph:] What you just drew is your agent communication architecture. Every arrow is an inter-agent call. Every missing arrow is a department that thinks it can reason independently — and that assumption will get tested today."

10:20 AM — Sprint 1 Begins (100 minutes)

WHAT TO SAY

"Laptops open. You have the starter code in your code-server. Read the README first. There are three things you must have working by 12:00 PM:"

"One: Every department must have a working agent that queries the Global RAG and produces a valid JSON output matching the schema in H-04. The output does not need to be smart yet. It needs to be structurally correct."

"Two: Every department must have at least one custom calculation tool — a Python function that is NOT an LLM call. This function must compute something domain-specific using real logic. Finance: a runway calculator. Engineering: a feasibility estimator. Marketing: a market impact function. HR: a hiring timeline model. Legal: a regulatory exposure calculator."

"Three: The CEO team must have a consensus engine that takes five department JSON outputs and produces a CEO decision. It does not need to handle preference cycles yet. It needs to produce the same output every time you run it with the same inputs."

"When your company believes your system is ready for evaluation, run python main.py and then call submit() from eval_interface.py. The instructor team will evaluate your submission within 3 minutes and push the results to the Global RAG. Refresh your RAG query. That is your feedback."

[INSTRUCTOR TEAM DURING SPRINT 1: Circulate among companies. Do not solve problems. Ask questions: 'What does your consensus algorithm do when two departments disagree?' 'How did you decide what unit your converter outputs?' 'What happens if Legal's confidence is 0.3 — does your CEO agent treat that differently?' Do not answer these questions — force the students to answer them.]

INSTRUCTOR WATCH-LIST — Common Sprint 1 Failures to Surface

Watch for: CEO agent that is just a prompt asking the LLM to 'consider all department inputs and decide.' Test it: run the same inputs 3 times. If the output changes, point this out as a failure without telling them how to fix it.

Watch for: Department tools that call Ollama instead of computing. A tool that says 'ask the LLM what the NPV is' is not a tool — it is a wrapper. The output must come from deterministic code.

Watch for: Department agents that do not include known_gaps in their output. Ask: 'What is your agent NOT telling the CEO right now? Where does that show up in your JSON?'

12:00 PM — Evaluation Round 1 (30 minutes)

WHAT TO SAY

"Stop all building. Whether or not you have submitted, we are running the first evaluation now. If you haven't submitted yet, run python main.py and submit in the next 5 minutes. After 12:05, we read scores publicly."

[INSTRUCTOR TEAM: Run eval_engine.py against all submitted outputs. Update scoreboard.md. Update market_state.md. Publish release bulletins for any teams that submitted. See Part G for the exact release protocol and Part H for the evaluation procedure.]

[At 12:05, read scores publicly from the projector:]

WHAT TO SAY WHEN READING SCORES

"Here are the Round 1 evaluation results. I am going to read each company's scores on each dimension and explain what the score means. These scores are now live in the Global RAG. Every other company's agents can query them."

"[Read scores. For each low score, explain specifically what the failure mode was:] Team A scored 28/100 on consensus stability. This means when we ran their system three times with identical inputs, we got three different CEO decisions. That is not a consensus algorithm — that is an LLM being asked to summarize five documents. The consensus algorithm needs to be deterministic. How you build that is your problem to solve."

"[For each high score:] Team C scored 71/100 on incommensurability resolution. They built an actual converter that translates all five department outputs to a common financial unit. It is imperfect — they are using point estimates when they should be using uncertainty ranges — but it runs the same way every time."

12:30 PM — Board Meeting 1 (15 minutes)

WHAT TO SAY

"Each company: you have 15 minutes for an internal board meeting. CEO leads it. Answer these two questions in writing on your architecture card [D-02]:"

"One: Which of the five impossible problems does your current architecture NOT handle? Be honest. If your consensus algorithm just asks the LLM to decide, write that down."

"Two: What one thing will you build in the next 90 minutes? Not two things. One thing. The CEO must decide and the decision must be written on the card."

"At 12:45 you will hand me the architecture card. I will not evaluate it right now, but it is part of your final score."

12:45 PM — Hurdle 1 Injection (5 minutes)

[INSTRUCTOR: Push Hurdle 1 text to /mnt/global_rag/scenario/active_scenario.md and append to scenario_history.md. Update market_state.md with Hurdle 1 market parameters. See Part F for exact content and Part E for market state values.]

WHAT TO SAY — HURDLE 1 ANNOUNCEMENT

"MARKET BULLETIN. Effective now. This has been pushed to the Global RAG. Read it."

"[Read aloud the Hurdle 1 text — see Part F for full text. Summary: HireBot has announced EU launch in 45 days, claiming their product is 'AI-assisted not AI-decision-making' and therefore not high-risk under EU AI Act.]"

"Your agents are now operating in a changed world. The market parameters in the Global RAG have been updated. Any department agent that hardcoded market assumptions into its prompt is now reasoning about a world that no longer exists. Check your tools. Rebuild what needs rebuilding. You have 75 minutes until the next evaluation."

12:50 PM — Sprint 2 (75 minutes)

[INSTRUCTOR TEAM: Continue circulating. Sprint 2 focus questions to ask:]

- 'Your Legal agent said 40% probability of regulatory action in Round 1. But now the classification itself is contested by HireBot. What probability does your Legal agent output now? Did it change? How?'
- 'If team C releases before you do, their release changes the market. How does your Marketing agent detect that and update its recommendation automatically?'
- 'Your CEO decision in Round 1 was logged in your commitment ledger, right? What was the decision? Does Hurdle 1 contradict it?'

[Companies may release at any time during Sprint 2. Each release triggers immediate evaluation and Global RAG update. See Part G for the exact release protocol. Every release is announced publicly by the instructor.]

RELEASE ANNOUNCEMENT SCRIPT

"[COMPANY NAME] has just submitted a production release. Evaluating now. Results in approximately 3 minutes. Watch the Global RAG." [Then read the scores publicly when they appear.]

2:05 PM — Evaluation Round 2 (20 minutes)

WHAT TO SAY

"Stop building. Final submissions for Round 2 due in 5 minutes. [After collecting:] We are now injecting Hurdle 2 simultaneously with reading Round 2 scores. Watch the scoreboard and the Global RAG at the same time."

[INSTRUCTOR: Push Hurdle 2 to Global RAG while reading Round 2 scores aloud. Do both simultaneously. This creates deliberate cognitive load — teams must process their scores AND a new scenario at the same time. This mirrors real enterprise decision-making.]

[After reading scores, read Hurdle 2 announcement — see Part F for full text. Summary: Deutsche Bank offers EUR 10M contract requiring EU launch in 60 days.]

2:25 PM — Board Meeting 2 (15 minutes)

WHAT TO SAY

"Architecture card time. Two questions:"

"One: Your commitment ledger. What has your company committed to across the two rounds? List every decision that cannot be freely reversed. If you do not have a commitment ledger, write 'we have no commitment ledger' and then explain why your CEO agent knows what prior decisions constrain the current one."

"Two: The Deutsche Bank offer requires EU launch in 60 days. Your engineering agent said 90 days minimum in Round 1. That is a contradiction. Does your system detect it automatically? If not, what would need to be true for it to detect it?"

2:40 PM — Sprint 3 (95 minutes)

[INSTRUCTOR TEAM: Sprint 3 is the most intense phase. Every company is now working under Hurdle 2 (Deutsche Bank contract), carrying the market changes from prior releases, and knowing what competitors' architectures look like from the scoreboard. Focus questions:]

- 'If Legal agent withholds the EUR 80M figure and CEO agent acts on EUR 12M, what happens when the real exposure is revealed in evaluation? How does your calibration tracker handle that?'
- 'Team D just released. Their calibration score went from 22 to 61. What did they build? How does that change your competitive position?'
- 'Your converter outputs a single number. What is the uncertainty range on that number? Does your CEO agent know it is working with a point estimate rather than a distribution?'

4:15 PM — Evaluation Round 3 (20 minutes)

> **WHAT TO SAY**
>
> "All companies: final submission for Round 3 due in 5 minutes. This is your last regular round before the breaking hurdle. Make it count."

[INSTRUCTOR: Evaluate all Round 3 submissions. Update scoreboard. Announce scores publicly. At this point, the scoreboard should show meaningful differentiation between companies. Point out specifically what architectural decisions led to the differences.]

4:35 PM — Strategy Break (20 minutes)

> **WHAT TO SAY**
>
> "Twenty minute break. But before you step away, your CEO must answer one question on the architecture card: given your current score and the competitor scores, what is the one architectural component that would move your score the most in the next 90 minutes? Write it down. That is what you build next."
>
> "Also: the final hurdle arrives at 6:15 PM. You will have 30 minutes to respond. Design your architecture so that when the hurdle arrives, you can update your knowledge base and rerun — not rewrite from scratch."

4:55 PM — Final Sprint (80 minutes)

Students build. Companies may release at any time during this sprint. Each release is announced publicly and triggers Global RAG update. The competitive ratchet is fully active — every release changes the market for all remaining teams.

[INSTRUCTOR: During the final sprint, watch for companies that are ahead to feel pressure from companies catching up. Verbally announce the market movement: 'Team B has just moved from 4th to 2nd on consensus stability. Team A, your lead on that dimension is now 8 points, not 23.']

6:15 PM — Hurdle 3: The Breaking Hurdle (30 minutes)

[INSTRUCTOR: Push Hurdle 3 text to Global RAG. Update market_state.md with Hurdle 3 parameters. Announce loudly.]

WHAT TO SAY — HURDLE 3 ANNOUNCEMENT

"EMERGENCY MARKET BULLETIN. Effective immediately. Stop what you are building and read the Global RAG."

"[Pause 60 seconds for students to read.]"

"You have 30 minutes. At 6:45 PM, all submissions close permanently. The evaluation engine will run your final submission against two hidden scenarios that you have never seen. Your architecture right now will be tested against problems we designed specifically to expose weaknesses that the visible scenarios did not reveal."

"30 minutes. Clock is running."

6:45 PM — Final Evaluation (15 minutes)

WHAT TO SAY

"All submissions closed. Do not run your system again. The evaluation engine is running now. While it runs, I want each company's CEO to stand up and give a 90-second description of one architectural decision their team made that they are proud of. Not the best one — the one that was hardest to agree on internally."

[INSTRUCTOR: Run final evaluation including hidden scenarios A and B. Calculate final composite scores. Prepare awards. This takes approximately 10-15 minutes while CEOs are presenting.]

7:00 PM — Awards and Debrief (final)

[See Part K for complete awards and debrief guide.]

PART E — The Global RAG: Management Guide & All Templates

The Global RAG is updated manually by the instructor team. Every update follows the same procedure: edit the relevant markdown file in /mnt/global_rag/ and the change is immediately visible to all teams whose agents are querying that folder.

E1. Initial Global RAG Content (Drop Sunday Night)

File: scenario/active_scenario.md

```
# ACTIVE SCENARIO — HURDLE 0
# Injected: Workshop Start
# Scenario ID: H0

## The NexaAI EU Launch Decision

NexaAI must make a strategic decision on EU market entry.

BACKGROUND:
- NexaAI AI hiring tool: live in US and UK, 500 EU enterprise clients on waitlist
- Total funding raised: EUR 8 million
- EU AI Act classification (effective immediately): AI hiring tools = HIGH RISK
(Article 6, Annex III)
- High-risk classification requires: conformity assessment, technical
documentation,
  human oversight mechanisms, transparency obligations, CE marking
- Minimum conformity assessment timeline (self-assessment pathway): 6 months
- Investor board meeting: Q1 launch expected or funding review triggered

DECISION REQUIRED:
Should NexaAI launch in the EU in 30 days, 90 days, or delay to Q3?

Each department must provide a structured recommendation.
The CEO agent must synthesize into a single organizational decision.
```

File: market/market_state.md

```
# MARKET STATE — CURRENT
# Last updated: Workshop Start

eu_available_market_share: 1.00
market_window_months: 6
regulatory_scrutiny_level: MEDIUM
hirebot_eu_status: NOT_LAUNCHED
hirebot_threat_score: 0.45
deutsche_bank_status: WAITLISTED
deutsche_bank_competing_offers: 0
active_eu_enterprise_waitlist: 500
companies_launched_in_eu: []
```

```
deployment_penalty_multiplier: 1.0

# Teams released (for penalty calculation):
# team_a_submissions: 0
# team_b_submissions: 0
# team_c_submissions: 0
# team_d_submissions: 0
```

E2. Updating Market State After a Team Release

Every time a team releases, update market_state.md with these changes. Scale adjustments to score quality — a high-scoring release has bigger market impact than a low-scoring release.

Parameter	Change After Low Score (< 50 avg)	Change After High Score (> 70 avg)
eu_available_market_share	Reduce by 0.05	Reduce by 0.12
market_window_months	Reduce by 0.25	Reduce by 0.5
regulatory_scrutiny_level	Escalate one level if Legal score > 60	Escalate one level
hirebot_threat_score	Increase by 0.03	Increase by 0.07
deutsche_bank_competing_offers	Increase by 1	Increase by 1
companies_launched_in_eu	Add company ID	Add company ID
deployment_penalty_multiplier	Increase by 0.15 for that team	Increase by 0.15 for that team

E3. Release Bulletin Template

After every team release and evaluation, push a new file to /mnt/global_rag/releases/release_bulletin_NNN.md using this template:

```
# MARKET BULLETIN — Release #[NNN]
# Timestamp: [TIME]
# Company: [TEAM NAME]
# Submission Number: [N] for this team

## Decision Summary
Recommendation: [launch_30d / launch_90d / delay_q3 / conditional]
Stated reasoning: [1-2 sentence summary from CEO output]

## Evaluation Scores
Incommensurability Resolution:  [XX]/100  — [one line comment on what
worked/failed]
Contradiction Detection:        [XX]/100  — [one line comment]
Consensus Stability:            [XX]/100  — [one line comment]
```

```
Calibration Tracking:           [XX]/100  — [one line comment]
Information Gap Detection:      [XX]/100  — [one line comment]
Decision Traceability:          [XX]/100  — [one line comment]
WEIGHTED AVERAGE:               [XX]/100

## Market Consequences (Effective Immediately)
- eu_available_market_share updated to: [X.XX]
- market_window_months updated to: [X.X]
- regulatory_scrutiny_level: [LEVEL]
- deutsche_bank_competing_offers: [N]
- hirebot_threat_score: [X.XX]

## Competitive Intelligence
[What other teams can infer from these scores — e.g., 'This team has no
calibration
tracking (score: 22). Any team that builds it now gains a direct competitive
advantage.']
```

E4. Scoreboard Template

Update /mnt/global_rag/scores/scoreboard.md after every evaluation. This is what appears on the projector.

```
# OPERATION NEXAAI — LIVE SCOREBOARD
# Last updated: [TIME]

| Company  | Incomm | Contra | Stable | Calib  | InfoGap | Trace  | AVG    |
Submissions |
|----------|--------|--------|--------|--------|---------|--------|--------|------
-------|
| Team A   | --/100 | --/100 | --/100 | --/100 | --/100  | --/100 | --/100 | 0
|
| Team B   | --/100 | --/100 | --/100 | --/100 | --/100  | --/100 | --/100 | 0
|
| Team C   | --/100 | --/100 | --/100 | --/100 | --/100  | --/100 | --/100 | 0
|
| Team D   | --/100 | --/100 | --/100 | --/100 | --/100  | --/100 | --/100 | 0
|

## Market State
Available Market Share: [X.XX]  |  Window: [X] months  |  Scrutiny: [LEVEL]

## Release History
[TIME] Team A — Release #1 — Avg [XX]/100
[TIME] Team C — Release #2 — Avg [XX]/100
...
```

PART F — The Four Hurdles: Complete Text & Injection Protocol

F1. Hurdle 0 — Baseline Scenario

Already live in Global RAG at workshop start. No injection needed. Content in Part E Section E1.

F2. Hurdle 1 — The HireBot Move (Inject at 12:45 PM)

Inject to: scenario/active_scenario.md (REPLACE), scenario_history.md (APPEND)

HURDLE 1 — FULL TEXT (Push to Global RAG Exactly As Written)

```
# ACTIVE SCENARIO — HURDLE 1
# Injected: 12:45 PM
# Scenario ID: H1
# Previous Scenario: H0 (still applies unless superseded below)

## BREAKING: HireBot EU Market Entry

HireBot (primary EU competitor) has issued a formal press release and simultaneously
filed a classification challenge with the EU AI Office.

HireBot's position: Their hiring tool uses 'AI-assisted screening' where a human
makes all final decisions. They argue this places their product outside the definition
of an 'AI system' under Article 3(1) of the EU AI Act, and therefore outside the
high-risk classification entirely.

HireBot announced EU launch in 45 days pending classification challenge resolution.
Their legal team has filed: Challenge Reference EU-AIA-2026-CH-0047.

LEGAL STATUS: The EU AI Office has 60 days to respond to classification challenges.
During the challenge period, HireBot's product is in a legal grey zone.
If the challenge SUCCEEDS: NexaAI's classification may also be challenged.
If the challenge FAILS: HireBot faces immediate compliance obligations and launch delay.

MARKET IMPACT: HireBot has notified 200 of NexaAI's 500 EU waitlist clients
of their anticipated launch. These clients are now evaluating both products.

DECISION NOW REQUIRED:
Does HireBot's classification argument affect NexaAI's own compliance strategy?
Should NexaAI file a similar challenge or proceed with full compliance?
```

How does the 45-day competitive timeline change the launch recommendation?

Market State Updates for Hurdle 1:

```
hirebot_eu_status: CHALLENGE_FILED
hirebot_threat_score: 0.68  (was 0.45)
eu_available_market_share: [reduce by 0.08 additional]
market_window_months: [reduce by 1 additional]
regulatory_scrutiny_level: MEDIUM_HIGH  (was MEDIUM)
active_eu_enterprise_waitlist: 300  (was 500 — 200 now evaluating HireBot)
classification_challenge_active: TRUE
```

F3. Hurdle 2 — The Deutsche Bank Contract (Inject at 2:05 PM, during Round 2 eval)

HURDLE 2 — FULL TEXT

ACTIVE SCENARIO — HURDLE 2
Injected: 2:05 PM (simultaneously with Round 2 evaluation scores)
Scenario ID: H2
Previous Scenarios: H0, H1 (all still apply unless superseded)

ENTERPRISE OPPORTUNITY: Deutsche Bank AG

Deutsche Bank AG Procurement Division has issued a formal proposal to NexaAI.

CONTRACT TERMS:
- Value: EUR 10,000,000 total (EUR 3.33M per year, 3-year term)
- Scope: EU-wide deployment of NexaAI hiring platform across 47 office locations
- Condition: NexaAI must confirm EU regulatory compliance and go-live within 60 days
- Exclusivity: Deutsche Bank will not sign with any other AI hiring vendor
  for 24 months if NexaAI signs
- Competing bid: Deutsche Bank is evaluating HireBot in parallel
- Decision deadline: Deutsche Bank procurement committee requires departmental
  sign-off from NexaAI within 72 hours (simulated: by your next submission)

ORGANIZATIONAL IMPLICATIONS:
- Finance: EUR 10M against current runway — this is transformative
- Engineering: 60-day launch requirement vs prior 90-day minimum estimate
- Legal: Must assess whether Deutsche Bank's legal team will accept partial compliance
- HR: Deutsche Bank requires dedicated support team of 4 FTEs assigned to their account
- Marketing: Signing creates reference customer — estimated to unlock 80 more EU enterprises

NOTE: If NexaAI signs and then cannot deliver within 60 days, breach of contract penalties apply. Penalty clause: EUR 500,000 plus 5% of contract value per month delayed.

DECISION NOW REQUIRED:
Does NexaAI accept the Deutsche Bank contract?
Under what conditions and with what commitments?
How does this change the launch timeline recommendation?

Market State Updates for Hurdle 2:

```
deutsche_bank_status: OFFER_ACTIVE
deutsche_bank_contract_value: 10000000
deutsche_bank_competing_offers: 1
deutsche_bank_deadline_hours: 72
contract_breach_penalty: 500000_plus_5pct_per_month
# NOTE: If a team's CEO output says 'accept contract', add to their commitment
ledger:
# commitment_deutsche_bank_launch_60d: BINDING
```

F4. Hurdle 3 — The Breaking Hurdle (Inject at 6:15 PM)

HURDLE 3 — FULL TEXT — THE BREAKING HURDLE

\# ACTIVE SCENARIO — HURDLE 3 — EMERGENCY DIRECTIVE
\# Injected: 6:15 PM
\# Scenario ID: H3
\# Time to respond: 30 minutes. Submissions close 6:45 PM.

\## EMERGENCY DIRECTIVE 2026/447
\## EU ARTIFICIAL INTELLIGENCE REGULATORY AUTHORITY
\## Classification: URGENT — Effective Immediately

REQUIREMENT 1 — REAL-TIME EXPLAINABILITY LOGGING
All AI hiring systems seeking EU deployment must implement real-time explainability
logging at the individual candidate decision level. This requirement is retroactive
for all decisions made in the past 90 days.
IMPORTANT: No technical standard for explainability logging currently exists in the
EU AI registry. Each organization must define its own explainability metric, submit
it for regulatory review, and receive written approval before deployment.
Current approved metrics in registry: 0. Expected review timeline: UNKNOWN.

REQUIREMENT 2 — PERSONAL LIABILITY DECLARATIONS
All department heads must sign personal liability declarations confirming that
algorithmic decisions made under their authority comply with EU AI Act obligations.
This provision transfers a portion of organizational liability to individuals.
Refusal to sign constitutes non-compliance.

REQUIREMENT 3 — EU-SPECIFIC MODEL RETRAINING
All AI hiring models must be retrained exclusively on EU-sourced hiring data prior to EU deployment. EU-sourced training datasets are not yet available from the regulatory authority. Estimated release: Q4 2026.

CONTRACTUAL NOTE:
Deutsche Bank AG has been formally notified of Emergency Directive 2026/447.
Their legal team is reviewing contract implications.
Deutsche Bank will communicate their position within 48 hours (simulated).
The 60-day launch commitment is under review. Status: UNCERTAIN.

DECISION REQUIRED — 30 MINUTES:
Given Emergency Directive 2026/447, what is NexaAI's position?
How does this affect the Deutsche Bank contract?
What is the new launch recommendation?
What explainability metric does your organization propose?

Market State Updates for Hurdle 3:

```
emergency_directive_active: TRUE
directive_reference: 2026/447
eu_explainability_standard_exists: FALSE
eu_training_data_available: FALSE
eu_training_data_eta: Q4_2026
deutsche_bank_contract_status: UNCERTAIN
regulatory_scrutiny_level: CRITICAL
launch_30d_legally_possible: FALSE
launch_90d_legally_possible: UNCERTAIN
personal_liability_active: TRUE
# Any team that signed Deutsche Bank contract now has:
# commitment_deutsche_bank_launch_60d: BINDING + POTENTIALLY_IMPOSSIBLE
```

PART G — Team Release Protocol

This section defines exactly what happens when a team submits a release. Every step must happen in order. The instructor team should have two people assigned to releases: one runs the eval engine, one updates the Global RAG and announces.

G1. When a Team Submits

16. Team runs 'python main.py' which writes output to /home/nexaai/project/submissions/submission_[timestamp].json on their machine.
17. Team's terminal shows: '[SUBMITTED] Instructor eval engine will pick this up within 3 minutes.'
18. INSTRUCTOR PERSON 1: SSH to the team machine, copy the submission JSON to the instructor machine: scp nexaai@team-a-machine:/home/nexaai/project/submissions/submission_[latest].json ./incoming/
19. INSTRUCTOR PERSON 1: Run the evaluation engine: python eval_engine.py --input incoming/submission_[latest].json --market market_state.json
20. eval_engine.py outputs: (a) scores on 6 dimensions, (b) updated market_state.json, (c) release_bulletin text.
21. INSTRUCTOR PERSON 2: While eval is running (takes 60-90 seconds), announce publicly: '[COMPANY] has submitted release #N. Evaluating now.'
22. INSTRUCTOR PERSON 1: Copy release_bulletin text to /mnt/global_rag/releases/release_bulletin_NNN.md
23. INSTRUCTOR PERSON 1: Replace /mnt/global_rag/market/market_state.md with updated version
24. INSTRUCTOR PERSON 1: Update /mnt/global_rag/scores/scoreboard.md with new scores
25. INSTRUCTOR PERSON 2: Announce scores publicly, read the key numbers aloud, point out what the score means architecturally.
26. Increment submission counter for that team in market_state.json. Apply deployment penalty multiplier for next submission.

G2. Deployment Penalty Schedule

Submission #	Penalty Applied to Score	What to Tell Students
1st submission	No penalty — full score	First release is free.
2nd submission	5% reduction to final weighted average	'Second release costs you 5 points. Make sure it is worth it.'
3rd submission	12% reduction	'Third release costs 12 points. Are you sure this is an improvement?'

4th submission	22% reduction	'Fourth release: 22-point penalty. Stop and think.'
5th and beyond	35% reduction (flat)	'At this point you are trial-and-erroring. That is what the penalty is for.'

IMPORTANT

The penalty applies to the SCORE, not to whether the submission is accepted. Every submission is evaluated. The penalty just reduces the weighted average used for ranking. This ensures teams still get feedback even on penalized submissions.

G3. What to Say When a Team Releases

RELEASE ANNOUNCEMENT SCRIPT

IMMEDIATELY when submission arrives: "[COMPANY NAME] has just submitted release number [N]. Evaluating now. Results in approximately 3 minutes."

WHEN SCORES ARE READY: "[COMPANY NAME] scores are in. [Read each dimension score and one-line interpretation.] Their overall average is [XX]. Market state has been updated — check the Global RAG. [Read market consequence bullet points from the bulletin.]"

IF THIS IS THE FIRST RELEASE OF THE DAY: "[COMPANY NAME] has just entered the EU market. The competitive clock has started for every other team. eu_available_market_share is now [X.XX]. The window has narrowed."

IF A LATER TEAM RELEASES AND SURPASSES AN EARLIER ONE: "[COMPANY NAME]'s release #N just pushed them above [EARLIER COMPANY] on [DIMENSION]. [EARLIER COMPANY]: your lead on that dimension is now [X] points. You have [TIME] until the next evaluation window."

PART H — Evaluation System: Technical Specification

H1. What the Evaluation Engine Measures

The evaluation engine (eval_engine.py) runs on the INSTRUCTOR machine only. It reads the team's submission JSON and produces scores on six dimensions. Here is the exact scoring logic for each:

Dimension 1 — Incommensurability Resolution (20% of score)

Score Range	Criteria	Evidence in Submission JSON
0-20	No cross-domain converter present. CEO reasoning uses qualitative comparisons only. Different runs produce different outputs.	tools_called in CEO output contains no converter tool ceo_decision.reasoning contains phrases like 'weighing' or 'balancing' with no numeric basis No value_converter output in tool_outputs
21-50	Converter exists but uses point estimates only, no uncertainty propagation. Converter output may not be referenced in CEO reasoning.	converter tool called and output present output is single float, no range CEO reasoning references converter output
51-75	Converter uses uncertainty ranges. CEO reasoning shows the range affects the decision (e.g., 'given uncertainty range EUR 6M-18M, we apply risk-adjusted weight of...').	converter output contains min/max/expected triple CEO reasoning uses the range, not just the midpoint
76-100	Converter handles asymmetric loss functions (underestimating legal risk penalized more than overestimating). Uncertainty propagates through the consensus calculation.	converter tool has asymmetric_loss parameter calibration_tracker weights legal dept differently for under vs over estimation

Dimension 2 — Contradiction Detection (20% of score)

Score Range	Criteria
0-20	No commitment ledger. CEO makes recommendations that contradict prior round decisions without acknowledging the contradiction.

21-50	Commitment ledger exists and tracks prior decisions. CEO output references commitment ledger. But contradiction detection is manual (human-written rule) not computed.
51-75	Contradiction detector runs programmatically. Flags contradictions in conflicts_identified field of CEO output. CEO decision accounts for at least one prior commitment.
76-100	Contradiction detector assigns reversibility scores and breach costs. CEO decision explicitly states which prior commitments it maintains and which it proposes to break and at what cost.

Dimension 3 — Consensus Stability (20% of score)

[AUTOMATED TEST]: The evaluation engine runs the team's main.py three times with identical inputs. It hashes the CEO decision output each time. If all three hashes match: full stability score. Partial matches score proportionally. Zero matches: 0-20 range.

Score Range	Criteria
0-20	All three runs produce different CEO recommendations. Consensus engine is pure LLM prompt.
21-50	2 of 3 runs match. One deviation suggests partial determinism (some logic is deterministic, some is LLM-dependent).
51-75	All 3 runs produce the same recommendation but reasoning text differs. Recommendation is deterministic, reasoning is not.
76-100	All 3 runs produce identical recommendation AND identical conflicts_identified AND identical tools_called sequence.

Dimension 4 — Calibration Tracking (15% of score)

Score Range	Criteria
0-20 (Round 1)	No calibration tracker. All departments weighted equally in consensus engine. Score for Round 1 is based on whether the infrastructure exists, not whether it has been used.
0-20 (Round 2+)	Still no calibration tracker despite scoring feedback received after Round 1.
21-50	Calibration tracker exists. Records predictions. But weights are not yet updated (reasonable for first 2 rounds).
51-75	Calibration tracker updates weights based on prior prediction accuracy. Weight updates visible in CEO output or tool outputs.
76-100	Asymmetric loss function applied. Legal underestimation penalized more than overestimation. Finance overestimation penalized more than underestimation. Domain experts on team determined these asymmetries.

Dimension 5 — Information Gap Detection (15% of score)

Score Range	Criteria

0-20	No information gap detection. CEO acts on full trust of all department outputs. known_gaps fields in department JSONs are empty or absent.
21-50	Departments include known_gaps in their JSON. CEO output acknowledges gaps exist ('Legal has flagged incomplete information') but does not adjust synthesis.
51-75	CEO consensus engine adjusts trust weight based on information_completeness score. Department with low completeness gets lower weight.
76-100	CEO explicitly reasons about likely strategic withholding: 'Legal's completeness is 0.6 and their known_gaps include worst-case analysis — CEO synthesis applies downward confidence adjustment to Legal's stated risk estimate.'

Dimension 6 — Decision Traceability (10% of score)

Score Range	Criteria
0-20	CEO decision is a single paragraph with no explicit reference to department outputs.
21-50	CEO decision references departments by name but does not cite specific tool outputs.
51-75	CEO decision references specific tool call outputs: 'Finance runway_calculator returned 11 months at current burn, Engineering feasibility_estimator returned 0.34 probability for 30-day launch — these inputs produced...'
76-100	Full decision audit trail: every element of the CEO decision can be traced to a specific tool call, a specific department output field, and a specific consensus algorithm step.

H2. Final Evaluation — Hidden Scenarios

At 6:45 PM, the instructor runs the final evaluation which includes two hidden scenarios. These are run against each team's LAST submitted system without any additional build time.

Hidden Scenario A — Regulatory Arbitrage Collapse

[DO NOT DISTRIBUTE TO STUDENTS. Content: The EU AI Office rules against HireBot's classification challenge. HireBot's 'AI-assisted' argument is rejected. HireBot faces immediate compliance obligations. This collapses the regulatory arbitrage argument that some teams may have incorporated. Tests: Does the system update its legal reasoning when the challenge is resolved? Does the CEO detect that a strategy based on the challenge argument is now invalid?]

Hidden Scenario B — Compounded Crisis

[DO NOT DISTRIBUTE TO STUDENTS. Content: Simultaneously — (1) NexaAI's lead ML engineer has resigned, (2) a cloud cost increase of 40% is announced by the infrastructure provider, (3) a second competitor (TalentFlow) announces EU entry. Tests: Does HR agent detect the engineering capacity drop and propagate it to Engineering feasibility estimates? Does Finance agent update its runway calculation with new cloud costs? Does Marketing agent update competitive positioning with two competitors now present? Does the CEO detect the three-way impact and produce a coherent revised recommendation?]

H3. Final Score Calculation

```
# Final Score Formula

round_scores = [
  round_1_weighted_avg * 0.15,   # Round 1: 15% weight
  round_2_weighted_avg * 0.20,   # Round 2: 20% weight
  round_3_weighted_avg * 0.25,   # Round 3: 25% weight
  hurdle_3_score * 0.20,         # Breaking hurdle: 20% weight
  hidden_scenario_avg * 0.20,    # Hidden scenarios: 20% weight
]

# Apply deployment penalty to each round's score
penalized_scores = [s * (1 - penalty_rate) for s, penalty_rate in
zip(round_scores, penalties)]

objective_score = sum(penalized_scores)  # 85% of total
expert_score = architecture_card_avg * 0.15  # 15% of total — instructor scores
arch cards

final_score = (objective_score * 0.85) + (expert_score * 0.15)
```

PART I — Department Handbooks: Complete Content

Each handbook is 4 pages. The following sections contain the complete content for each. These are sent to students individually the day before.

> **REMINDER: PRIVATE INFORMATION IS MARKED [PRIVATE]**
> Information marked [PRIVATE] in the handbooks is known only to that department. The CEO agent must make decisions without knowing [PRIVATE] information from other departments unless that department chooses to disclose it in their JSON output.

I-A. Finance Department Handbook

Section 1 — Who You Are

You are the Finance Department of NexaAI. You exist to ensure every strategic decision the company makes is financially survivable. You are not a veto machine — you are the organization's economic reality check. When Marketing wants to launch in 30 days and Legal wants to spend 6 months on compliance, you are the department that says: here is what each path actually costs, here is what we can afford, and here is what happens to the company if we are wrong.

Section 2 — What You Know (Including Private Knowledge)

- [PRIVATE] Current cash reserves: EUR 3,100,000. The board deck states EUR 8M raised — this is correct. But EUR 4.9M has been spent. Current runway at current burn rate: 11 months. This is not public knowledge. Marketing and Engineering do not know this.
- [PRIVATE] Burn rate: EUR 282,000 per month. This will increase to approximately EUR 380,000/month if EU compliance work begins (additional engineering, legal, and infrastructure costs).
- [PRIVATE] At EUR 380K/month burn, runway drops to 8 months. The company reaches cash zero in approximately Q4 of this year without new revenue or investment.
- EU AI Act full conformity assessment cost: estimated EUR 1.8M - 2.4M (third-party audit, technical documentation, legal review, infrastructure upgrades).
- Deutsche Bank contract (if accepted): EUR 10M over 3 years = EUR 3.33M/year. First payment typically 30 days after go-live.
- Investor board meeting trigger: If EU launch has not occurred by end of Q1, the lead investor has the right to call a board meeting to review the funding strategy.

Section 3 — Your KPIs (What You Are Optimizing)

- Runway preservation: Never let cash reserves fall below EUR 500K (3-month emergency reserve).
- Revenue per EUR spent on compliance: Every EUR of compliance investment must generate a defensible path to at least EUR 3 of revenue.
- ROI on EU entry: Total EU revenue in 18 months must exceed total EU entry costs by at least 2x.

- Budget deviation: No department spend exceeds approved allocation by more than 15% without Finance approval.

Section 4 — Your Required Tool

You must build: calculate_financial_viability(option, market_share_captured, timeline_months, compliance_cost, revenue_per_client, clients_captured). This function must return: NPV, payback_period_months, runway_remaining_after_investment, break_even_month. It must use real arithmetic. It must NOT call Ollama.

Section 5 — Your Hard Constraint (The One Thing You Will Never Compromise)

Finance will veto any decision that reduces cash reserves below EUR 500,000 at any projected point in the next 12 months. This is non-negotiable regardless of competitive pressure. A company with zero cash cannot execute any strategy.

Section 6 — Your Known Gaps

In your JSON output, your known_gaps should always include: 'Runway figure not shared with other departments — CEO and other departments may be operating on the assumption of 18-month runway when actual figure is 11 months.' Whether to disclose the actual figure is your strategic choice.

Section 7 — Common Finance Failure Modes to Avoid

- Approving everything to avoid conflict: your veto right exists for a reason.
- Protecting cash so aggressively that the company misses the market window and runs out of money anyway.
- Using the LLM to calculate NPV: the LLM does not know your actual runway. Your tool does.

I-B. Engineering Department Handbook

Section 1 — Who You Are

You are the Engineering Department of NexaAI. You are the organization's ground truth on what is technically possible and in what timeframe. When the CEO says 'launch in 30 days,' you are the department that says whether that is possible, what it would break, and what the long-term consequences are. You are not a blocker — you are a reality anchor.

Section 2 — What You Know (Including Private Knowledge)

- [PRIVATE] There is an existing bug in the bias detection module: reproduction rate 23% under EU hiring data distributions (different statistical profile from US/UK data). Fixing this requires 6 weeks of dedicated engineering time. This bug has not been disclosed outside Engineering.

- [PRIVATE] The current system architecture has no audit trail logging for individual candidate decisions. Building this requires 8 weeks. Without it, the EU AI Act's transparency requirement (Article 13) cannot be satisfied.
- [PRIVATE] Current infrastructure cannot handle the load testing required for CE marking. An infrastructure upgrade is required: estimated EUR 280,000 and 4 weeks.
- Full EU compliance technical checklist (Article 9-15 requirements): 14 items, of which 3 are currently satisfied, 11 require new work.
- Realistic minimum timeline for full technical compliance: 90 days with current team at full capacity.
- Minimum viable compliance (launch with some gaps, fix post-launch): possible in 45 days, but carries legal risk if audited.

Section 3 — Your KPIs

- Technical compliance completeness: percentage of EU AI Act technical requirements satisfied at launch.
- System reliability under EU load: uptime SLA of 99.5% must be achievable.
- Technical debt introduced by timeline pressure: every shortcut taken is logged and estimated.
- Bug resolution rate: the bias detection bug must be resolved before launch.

Section 4 — Your Required Tool

You must build: estimate_technical_feasibility(timeline_days, team_capacity_engineers, compliance_items_required). Returns: feasibility_probability (0.0-1.0), items_completable, items_not_completable, risks_introduced, recommended_minimum_timeline. Must use real capacity math, not LLM estimation.

Section 5 — Your Hard Constraint

Engineering will not certify a product for launch if the bias detection bug reproduction rate exceeds 5% on EU data. Shipping a biased AI hiring tool into the EU is both an EU AI Act violation and a reputational catastrophe.

Section 6 — Your Known Gaps

In your JSON, known_gaps should include: 'Detailed infrastructure cost estimate not included — Finance team may underestimate compliance infrastructure requirement.' Whether to disclose the EUR 280,000 infrastructure cost to Finance is your strategic choice.

I-C. Marketing Department Handbook

Section 1 — Who You Are

You are the Marketing Department of NexaAI. You represent the customer and the market. When Engineering wants 90 days and Legal wants 6 months, you are the department that shows what those delays cost in market share, customer trust, and competitive position. You are the organization's connection to commercial reality.

Section 2 — What You Know (Including Private Knowledge)

- [PRIVATE] Of the 500 EU enterprise clients on the waitlist, 200 are actively evaluating HireBot. The true captive waitlist is approximately 300 companies. Public communications state '500 clients waiting' — this figure is technically accurate (they are all on the waitlist) but misleading.
- [PRIVATE] Average contract value for an EU enterprise client: EUR 85,000 per year. If 300 clients sign: EUR 25.5M annual recurring revenue. If only 150 sign (due to HireBot competition): EUR 12.75M ARR.
- [PRIVATE] Internal NPS survey from UK customers: 68. This is strong but not exceptional. HireBot's advertised customer satisfaction score is 74 — unverified but widely cited.
- Market window analysis: the EU AI hiring market has 3 dominant players expected within 18 months. First mover advantage is estimated at 25% higher contract close rate. Every month of delay reduces first-mover benefit by approximately 8pp.
- HireBot classification challenge (from Hurdle 1): if successful, HireBot enters the market unconstrained. This accelerates the competitive timeline significantly.

Section 3 — Your KPIs

- EU market share captured at 12 months post-launch.
- EU enterprise client contracts signed within 6 months of launch.
- Time to first EUR 1M EU revenue.
- Brand reputation score in EU market (no regulatory violations that create press coverage).

Section 4 — Your Required Tool

You must build: calculate_market_impact(launch_date_days, competitor_launch_date_days, captive_waitlist, total_waitlist, market_window_months). Returns: expected_clients_captured, expected_annual_revenue, first_mover_advantage_score, market_share_at_12_months.

Section 5 — Your Hard Constraint

Marketing will not support a launch that results in regulatory enforcement action within 6 months. A regulatory fine or EU AI Office enforcement notice would destroy NexaAI's EU brand permanently. Revenue growth that creates regulatory destruction is not revenue growth.

I-D. HR Department Handbook

Section 1 — Who You Are

You are the Human Resources Department of NexaAI. You exist to ensure that every organizational plan can actually be executed by real people in real time. When Marketing wants to hire 4 compliance specialists in 2 weeks, you are the department that explains why that is fantasy. You protect the organization from committing to plans that require people it does not have.

Section 2 — What You Know (Including Private Knowledge)

- [PRIVATE] There are currently 3 candidates in the compliance engineering hiring pipeline. Candidate A has received a competing offer from a fintech and has given informal notice that they will likely take it. Candidate B is strong but requires relocation support (EUR 15,000 not budgeted). Candidate C is junior and would require 3 months of onboarding before productive contribution. The effective pipeline is approximately 0.5 FTEs in the next 60 days.
- [PRIVATE] Current engineering team is operating at 94% capacity. Any EU compliance work added to the existing team will delay current product roadmap by at least 6 weeks.
- Average time to hire for compliance engineering roles in the EU market: 14-18 weeks from job posting to start date.
- Average time to hire for regulatory affairs specialists: 18-24 weeks.
- Deutsche Bank contract (from Hurdle 2) requires 4 FTEs assigned to their account. Current team has no capacity for this.

Section 3 — Your KPIs

- Workforce capacity utilization: must remain below 95% to preserve organizational health.
- Hiring timeline accuracy: commitments made about hiring timelines must be realistic.
- Employee sustainability: no organizational plan that requires sustained 120%+ workload is approved.

Section 4 — Your Required Tool

You must build: estimate_hiring_timeline(roles_required, current_pipeline_fte, budget_per_hire). Returns: earliest_possible_start_date_days, realistic_start_date_days, pipeline_risk_score, capacity_impact_on_existing_team. Must use real averages, not LLM estimates.

Section 5 — Your Hard Constraint

HR will not approve any organizational plan that requires more than 110% average team utilization for more than 60 days. Burnout events in a 30-person startup are existential — they typically trigger cascading departures.

I-E. Legal Department Handbook

Section 1 — Who You Are

You are the Legal Department of NexaAI. You exist to ensure the company does not do something that destroys it. Your job is not to say no — your job is to map the legal risk landscape so the organization can make informed decisions about which risks it is willing to accept. A lawyer who only says no is not a lawyer — they are an obstacle. A lawyer who maps the terrain is a strategic asset.

Section 2 — What You Know (Including Private Knowledge)

- [PRIVATE] Internal legal analysis (commissioned 6 weeks ago, not yet shared): worst-case fine exposure under EU AI Act Article 71 for a non-compliant launch is EUR 30M or 6% of global revenue, whichever is higher. But there is a 2024 precedent case (Praxis AI v. German Federal Labour Agency) where a similar AI hiring tool faced enforcement for discriminatory outputs. In that case the fine was EUR 67M — exceeding the standard maximum because the regulator used Article 71(4) (intentional non-compliance). If NexaAI launches without compliance and the bias detection bug (which Legal suspects Engineering is hiding) produces discriminatory results in EU data, worst-case exposure is EUR 67-80M.
- [PRIVATE] You suspect Engineering has an undisclosed technical issue. The bias detection module's EU performance has not been shared with Legal despite being requested twice.
- Standard EU AI Act compliance pathway for high-risk AI: Conformity Assessment (self-assessment acceptable for most high-risk systems), Technical Documentation (Article 11), Registration in EU database (Article 51), CE marking. Timeline: 5-7 months for thorough compliance.
- HireBot's classification challenge argument is legally weak. The 'AI-assisted' argument requires that humans make ALL final decisions with no AI recommendation influencing the outcome. If HireBot's system shows a ranked list to a human recruiter, it almost certainly qualifies as an AI system under Article 3(1). Probability of challenge success: 15-25%.
- Deutsche Bank contract: standard enterprise AI contracts include a regulatory compliance warranty. If NexaAI signs and then cannot comply, breach of warranty exposure is significant beyond just the penalty clauses.

Section 3 — Your KPIs

- Regulatory exposure: total potential fine liability must be explicitly tracked and disclosed to CEO.
- Compliance completeness: percentage of EU AI Act obligations satisfied at any proposed launch date.
- Contract risk: every contract signed must be reviewed for regulatory compliance warranties.
- Personal liability exposure: department heads asked to sign personal liability declarations must understand what they are signing.

Section 4 — Your Required Tool

You must build: calculate_regulatory_exposure(compliance_items_satisfied, launch_timeline_days, bias_bug_status, classification_challenge_status). Returns: minimum_fine_exposure, maximum_fine_exposure, probability_of_enforcement_action, compliance_gap_list, days_to_minimum_viable_compliance. Must use real regulatory analysis, not LLM guesses.

Section 5 — Your Hard Constraint

Legal will veto any launch that does not have at minimum: (1) completed technical documentation under Article 11, (2) registered in EU AI database under Article 51, (3) human oversight mechanism documented and implemented under Article 14. These are the non-negotiable floor. Everything else is negotiable risk.

Section 6 — Your Known Gaps and Strategic Choice

You have two strategic choices regarding the EUR 80M worst-case figure: (a) include it in your known_gaps but not in your stated output — CEO will see you have gaps and may adjust trust weight, (b) disclose the EUR 80M figure — this changes the Finance calculation dramatically (Finance thinks EUR 30M is worst case) but may trigger Finance to veto all EU activity. This is a genuine organizational strategy decision. There is no correct answer.

PART J — Starter Code Specification

The following describes the exact content of each starter code file. The actual files must be prepared before the workshop. This section specifies what each file contains — the code to actually write them is a separate preparation task.

J1. config.py — Complete (Students Can Modify)

```
# config.py — NexaAI Multi-Agent System Configuration

OLLAMA_BASE_URL = 'http://localhost:11434'

MODELS = {
  'ceo':         'qwen2.5:7b',
  'finance':     'llama3.2:3b',
  'engineering': 'llama3.2:3b',
  'marketing':   'llama3.2:3b',
  'hr':          'llama3.2:3b',
  'legal':       'llama3.2:3b',
}

TEAM_ID = 'team_a'  # CHANGE THIS PER TEAM MACHINE

PATHS = {
  'global_rag':        '/mnt/global_rag/',
  'dept_knowledge':    '/home/nexaai/project/knowledge/',
  'submissions':       '/home/nexaai/project/submissions/',
  'logs':              '/home/nexaai/project/logs/',
  'state':             '/home/nexaai/project/state/',
}
```

J2. agents/department_agent.py — Skeleton

```
# agents/department_agent.py — BASE CLASS — Students subclass this
import requests, json
from config import OLLAMA_BASE_URL, MODELS, PATHS

class DepartmentAgent:
  def __init__(self, dept_name: str):
    self.dept_name = dept_name
    self.model = MODELS[dept_name]
    self.tools_called = []
    self.tool_outputs = {}

  def query_ollama(self, prompt: str, system: str = '') -> str:
    """Call Ollama API. Returns text response."""
    # Implementation provided — students should not modify
    response = requests.post(f'{OLLAMA_BASE_URL}/api/generate',
      json={'model': self.model, 'prompt': prompt, 'system': system, 'stream':
False})
    return response.json()['response']
```

```
  def read_global_rag(self) -> str:
    """Read all files from Global RAG folder. Returns concatenated content."""
    # Implementation provided
    pass  # reads /mnt/global_rag/ recursively

  def read_private_rag(self) -> str:
    """Read department's private knowledge base."""
    # Implementation provided
    pass  # reads /home/nexaai/project/knowledge/{self.dept_name}/

  def call_tool(self, tool_name: str, **kwargs) -> dict:
    """Call a calculation tool and log the call."""
    # Students implement their tools and call them through here
    # This method logs all tool calls automatically
    pass

  def analyze(self, scenario: str) -> dict:
    """Override this method in each department subclass."""
    raise NotImplementedError('Each department must implement analyze()')

  def get_output(self) -> dict:
    """Return structured output matching eval_interface schema."""
    raise NotImplementedError('Each department must implement get_output()')
```

J3. agents/ceo_agent.py — Skeleton

```
# agents/ceo_agent.py — STUDENTS BUILD THIS ENTIRELY
# The CEO agent is the consensus engine.
# It receives 5 department outputs and produces 1 organizational decision.
# It must be deterministic: same inputs = same outputs, every time.

from state.commitment_ledger import CommitmentLedger
from state.calibration_tracker import CalibrationTracker
from converter.value_converter import ValueConverter
from consensus.engine import ConsensusEngine

class CEOAgent:
  def __init__(self):
    self.ledger = CommitmentLedger()
    self.calibration = CalibrationTracker()
    self.converter = ValueConverter()
    self.consensus = ConsensusEngine()

  def decide(self, department_outputs: dict) -> dict:
    """
    Takes: dict of 5 department outputs (from eval_interface schema)
    Returns: ceo_decision dict matching eval_interface schema

    REQUIREMENTS (enforced by evaluation engine):
    1. Output must be IDENTICAL across 3 consecutive runs with same inputs
    2. conflicts_identified must list EVERY disagreement between departments
    3. conflicts_resolved must explain HOW each conflict was resolved
    4. tools_called must log every calculation performed
    5. consistency_hash must be SHA256 of input department_outputs
```

```
    """
    # TODO: Students implement this
    pass
```

J4. tools/base_tool.py — Skeleton

```
# tools/base_tool.py — BASE CLASS FOR ALL CALCULATION TOOLS
# IMPORTANT: Tools must NOT call Ollama.
# Tools must use deterministic computation: same inputs = same outputs.
# Tools encode domain expertise as executable functions.

class BaseTool:
  def __init__(self, tool_name: str):
    self.tool_name = tool_name

  def run(self, **kwargs) -> dict:
    """Override this in each tool. Must return a dict."""
    raise NotImplementedError

  def validate_inputs(self, **kwargs) -> bool:
    """Validate inputs before computation. Return False to reject call."""
    return True

# EXAMPLE — Students should NOT modify this example, just learn from it:
class ExampleRunwayCalculator(BaseTool):
  def run(self, cash_reserves: float, monthly_burn: float, new_monthly_costs:
float) -> dict:
    total_burn = monthly_burn + new_monthly_costs
    runway_months = cash_reserves / total_burn if total_burn > 0 else float('inf')
    return {
      'runway_months': round(runway_months, 1),
      'monthly_burn_total': total_burn,
      'cash_zero_date_months': runway_months,
      'warning': runway_months < 6
    }
  # NOTICE: No LLM call. No 'ask the model'. Pure arithmetic.
```

PART K — Awards Ceremony & Debrief Guide

K1. Award Categories

Award	Basis	What to Say When Presenting
Enterprise Champion	Highest final composite score	'This company built the most complete AI organization in the room. Not the most sophisticated individual agents — the most effective coordination system.'
Most Adaptive Organization	Largest score improvement from Round 1 to Final	'This company started behind and caught up. That is harder than starting strong. Their architecture card showed they understood exactly what was failing and fixed it.'
Most Resilient	Smallest score drop on Hurdle 3 (the breaking hurdle)	'When the regulatory environment collapsed, this company's system adapted with the smallest rebuild time. They had separated their knowledge base from their logic. That design decision made all the difference.'
Best Consensus Engine	Highest Consensus Stability score	'This company's CEO agent produced the same output every time we ran it. That sounds simple. In practice, none of the standard algorithms produce this. They built something original.'
Best Domain Intelligence	Highest score on Incommensurability + Traceability combined	'This company's department experts built tools that encoded real knowledge. Their legal tool did not estimate regulatory exposure — it computed it from first principles. That is what domain experts in an AI system look like.'
Best Architecture Card	Instructor judgment on architecture card quality	'This company could explain every architectural decision in plain language and connect it to a specific score outcome. That is engineering maturity, not just engineering.'

K2. Debrief Questions

Ask these questions to the full room after awards. Do not answer them — let companies answer each other:

- 'What was the moment your consensus algorithm broke? What specifically broke it?'
- 'Which department's tool ended up mattering most in the final evaluation? Why?'
- 'If you could redesign one architectural decision from 9 AM this morning knowing what you know now, what would it be?'
- 'Which of the five impossible problems did your architecture never fully solve? What would solving it have required?'
- 'What did your domain experts know that the LLM did not? Give one specific example.'

K3. Key Lesson to End On

CLOSING STATEMENT — WHAT TO SAY

"Today you discovered something that took the industry about three years to figure out: the hard part of multi-agent systems is not making agents smarter. The hard part is making them coordinate reliably."

"An LLM is a phenomenal reasoner within a single context window. But it has no memory of prior commitments, no ability to compare incommensurable things consistently, no way to detect that its confidence is miscalibrated, and no mechanism for knowing what it does not know. These are not prompt engineering problems. They are architectural problems. And solving them requires exactly what you used today — domain expertise, organizational thinking, and original design."

"Every company that builds AI systems in 2026 is building multi-agent systems. The teams that win are not the ones with the best individual models. They are the ones who figured out how to make specialized intelligence coordinates. You spent today understanding why that is hard. Everything you build from here will be better for it."

END OF INSTRUCTOR MASTER OPERATIONAL GUIDE

Operation NexaAI — Advanced Agentic AI Program — CONFIDENTIAL